WORKED SOLUTION

STEP III 2022, Question 12

Probability and Statistics • Random Chords in a Circle

STEP III 2022, Question 12

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2022 STEP papers (a zip file):

www.ocr.org.uk/Images/696587-2022-past-papers.zip

This question has a full guided treatment in *Beyond STEP*, Section 12.3, *What Does "Random" Mean?*: https://beyondstep.keithmcnulty.org/ch12/bertrand.html

The key idea

Every chord in this question comes from one fact: a chord at perpendicular distance d from the centre of a circle of radius a has length $2\sqrt{a^2-d^2}$. So the work is to find d in terms of the random quantities, then average. In part (ii) the substitution $u = a^2 - r^2\sin^2 t$ is natural because the density of R supplies exactly the factor r that the substitution needs.

Solution

Part (i)

Let Θ be the angle AOB, so Θ is uniform on $[0, 2\pi]$ with density $\frac{1}{2\pi}$. Triangle AOB is isosceles with apex angle Θ. Dropping the perpendicular from O to AB bisects both the angle and the chord, so

$$AB = 2a\sin\tfrac{1}{2}\Theta.$$

This holds for all $\Theta \in [0, 2\pi]$, since $\sin\frac{1}{2}\Theta \geq 0$ there; for $\Theta > \pi$ we are measuring the reflex angle, but the chord is the same. Hence

$$\mathrm{E}(AB) = \frac{1}{2\pi}\int_0^{2\pi} 2a\sin\tfrac{1}{2}\theta\,\mathrm{d}\theta = \frac{a}{\pi}\Big[-2\cos\tfrac{1}{2}\theta\Big]_0^{2\pi} = \frac{a}{\pi}(2+2) = \frac{4a}{\pi}.$$

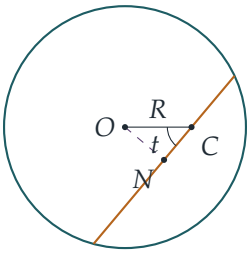
Part (ii)

Density of R. For $0 \le r \le a$, the event $R \le r$ means that C lies in the disc of radius r about O. Since probability is proportional to area,

$$\mathrm{P}(R \le r) = \frac{\pi r^2}{\pi a^2} = \frac{r^2}{a^2}, \qquad \text{so} \qquad f_R(r) = \frac{2r}{a^2} \quad (0 \le r \le a),$$

and $f_R(r) = 0$ otherwise.

Length of the chord. Let N be the foot of the perpendicular from O to the chord. In the right-angled triangle ONC, the hypotenuse is $OC = R$ and the angle at C is t, so $ON = R\sin t$.



The perpendicular from the centre bisects the chord, and N is its midpoint, so by Pythagoras in the triangle formed by O, N and an end of the chord,

$$\text{chord length} = 2\sqrt{a^2 - R^2\sin^2 t}.$$

Since $R < a$, the expression under the root is positive. Note that the whole chord is counted, on both sides of C.

The expected length. With t fixed and C random,

$$\mathrm{L}(t) = \int_0^a 2\sqrt{a^2 - r^2\sin^2 t}\,\frac{2r}{a^2}\,\mathrm{d}r.$$

Take $0 < t < \frac{1}{2}\pi$, so that $\sin t \neq 0$. Substitute $u = a^2 - r^2\sin^2 t$, so $\mathrm{d}u = -2r\sin^2 t\,\mathrm{d}r$; when $r = 0$, $u = a^2$, and when $r = a$, $u = a^2(1 - \sin^2 t) = a^2\cos^2 t$. Then

$$\mathrm{L}(t) = \frac{2}{a^2\sin^2 t}\int_{a^2\cos^2 t}^{a^2} u^{1/2}\,\mathrm{d}u = \frac{2}{a^2\sin^2 t}\cdot\frac{2}{3}\left(a^3 - a^3\cos^3 t\right) = \frac{4a(1-\cos^3 t)}{3\sin^2 t},$$

using $(a^2\cos^2 t)^{3/2} = a^3\cos^3 t$, which is valid because $\cos t \ge 0$ for an acute angle. This is the required formula.

Ask yourself How can the denominator $\sin^2 t$ be turned into something that cancels with $1 - \cos^3 t$?

Write $c = \cos t$. Then $\sin^2 t = 1 - c^2 = (1-c)(1+c)$, and $1 - c^3$ is a difference of cubes, $(1-c)(1+c+c^2)$. Since $c \neq 1$, we can cancel $1 - c$:

$$\mathrm{L}(t) = \frac{4a}{3}\cdot\frac{1+c+c^2}{1+c} = \frac{4a}{3}\left(c + \frac{1}{1+c}\right),$$

because $c(1+c)+1 = 1+c+c^2$. Finally, $1+\cos t = 2\cos^2 \frac{1}{2}t$, so $\dfrac{1}{1+\cos t} = \frac{1}{2}\sec^2 \frac{1}{2}t$, and

$$\mathrm{L}(t) = \frac{4a}{3}\Big(\cos t + \tfrac{1}{2}\sec^2\big(\tfrac{1}{2}t\big)\Big),$$

as required. This form has no trouble at $t = 0$: it gives $\frac{4a}{3}\cdot\frac{3}{2} = 2a$, which is right, since a chord through C along OC is a diameter.

Part (iii)

T has density $\frac{2}{\pi}$ on $[0, \frac{1}{2}\pi]$, so, using the second form of L (which is easy to integrate, since $\frac{\mathrm{d}}{\mathrm{d}t}\tan\frac{1}{2}t = \frac{1}{2}\sec^2\frac{1}{2}t$),

$$\begin{aligned}\mathrm{E}\big(\mathrm{L}(T)\big) &= \frac{2}{\pi}\cdot\frac{4a}{3}\int_0^{\pi/2}\Big(\cos t + \tfrac{1}{2}\sec^2\tfrac{1}{2}t\Big)\,\mathrm{d}t = \frac{8a}{3\pi}\Big[\sin t + \tan\tfrac{1}{2}t\Big]_0^{\pi/2}\\ &= \frac{8a}{3\pi}(1+1) = \frac{16a}{3\pi}.\end{aligned}$$

Checks

- At $t = \frac{1}{2}\pi$ the chord is perpendicular to OC, and $\mathrm{L} = \frac{4a}{3}(0 + \frac{1}{2}\cdot 2) = \frac{4a}{3}$. Directly, $\int_0^a 2\sqrt{a^2-r^2}\,\frac{2r}{a^2}\,\mathrm{d}r = \frac{4a}{3}$ as well.
- $\mathrm{L}(t)$ should fall from $2a$ to $\frac{4}{3}a$ as t goes from 0 to $\frac{1}{2}\pi$, and the answer to (iii), $\frac{16}{3\pi}a \approx 1.698a$, lies between these, as an average must.
- With $a = 1.7$ and $t = 0.7$, numerical integration of the chord length against the density of R gives 3.01799, matching both formulae for $\mathrm{L}(0.7)$. The answer to (i), $\frac{4}{\pi}a \approx 1.273a$, was also confirmed by numerical integration.

Common slips

- Taking R to be uniform on $[0, a]$. Points are uniform by *area*, and there is more area far from the centre, hence the density $2r/a^2$.
- Using only the part of the chord on one side of C, or writing $ON = R\cos t$ (that is the distance CN along the chord).
- Integrating the first form of $\mathrm{L}(t)$ in part (iii) instead of the second form, which the question has set up to integrate in one line.

Beyond STEP

This is geometric probability, where "choose at random" only means something once a distribution is fixed. The answers $\frac{4a}{\pi}$ and $\frac{16a}{3\pi}$ differ because they describe different ways of picking a chord, the point behind Bertrand's paradox. Integral geometry makes this

systematic, using measures on lines that do not change under rotation and translation. For continuous distributions on regions and expectations of the kind computed here, see Blitzstein and Hwang, *Introduction to Probability*.

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